Nexora Software Requirements Specification

Undergraduate Software Engineering Project - Professor Farrokhian

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Chapter 1

Purpose and Basis of Use

This document gives everyone who works with Nexora a common language, from the product owner to development, testing, and operations. Its goal is to give everyone the same answer to the question, “What should this system do?” Developers should be able to identify the exact expected behavior from each requirement. Testers should be able to check that behavior with defined inputs and determine whether it was implemented correctly.

This specification defines what is required and why. The technical design, installation and deployment guide, and test execution report are separate documents.

1.1 Document Structure and Reference

The structure follows requirements engineering topics in ISO/IEC/IEEE 29148:2018. The standard informed the framework. No formal certification or claim of conformity is made.

Each requirement has an identifier that remains stable throughout the life of the document, together with a description, acceptance criteria, priority, and reference.

Two expressions appear throughout this document:

• “Shall” identifies a requirement that must be met for acceptance.

• “Proposed target” identifies a value that still requires formal approval.

> Note: An identifier beginning with AT refers to a proposed test specification. An automated test file may not yet exist. For each test, the test management system shall record the result, software version, execution environment, date, and evidence so the outcome can be reconstructed.

1.2 Product Basis and References

Where did this specification come from?

It draws on two sources:

The recorded requests for Nexora and the repository code at the time of review.

The requirements therefore come from actual product needs or from behavior reflected in the code.

Two reference types appear throughout:

• C points to code that underlies the requirement.

• U points to user or business needs for the product.

Differences between documented and implemented behavior are collected under open decisions in Chapter 27. This distinguishes the target behavior from the current code and identifies issues that still require a decision.

Chapter 2

Product Scope and Stakeholders

Nexora is a web interface for using organizational knowledge. Administrators connect authorized documents and sources to knowledge sets. Background processing extracts text and makes it retrievable. Users ask questions within their authorized scope and see the answer, its sources, and its grounding status.

2.1 Scope of This Version

This version covers authentication, member management, sets, file upload and security checks, optional OCR, chunking, search, conversations, dedicated assistants, conversation sharing, feedback, analytics, notifications, and service operations. Payments, subscription purchases, native mobile apps, voice calls, automated execution of organizational decisions, and foundation-model training are outside its commitments.

2.2 Stakeholders and Responsibilities

Stakeholder	Need or responsibility
Organization user	Find answers with clear access to evidence and history.
Organization administrator	Manage members, documents, and sources within the organization.
Product owner	Approve scope, priorities, constraints, and acceptance criteria.
Test team	Check normal behavior, access boundaries, errors, and answer quality.
Security lead	Approve external providers, data retention, and secrets policy.

Operations lead	Deploy, monitor, back up, and recover the service.

2.3 Product Success Criteria

An answer shall stay within the selected scope, provide visible sources, and clearly state when evidence is insufficient. Administrators shall be able to diagnose why a document is not ready without relying on ambiguous progress percentages.

Chapter 3

Operational Context and Dependencies

The React and Vite interface communicates with a FastAPI API. PostgreSQL stores operational data and Qdrant stores retrieval vectors. Original files reside in document storage. In development, the frontend typically uses port 5173 and the API port 8000. Production deployments are free to use other ports.

This version was technically reviewed on 19 Mehr 1405 (October 11, 2026). The frontend moved to frontend/app, and its feature areas are organized under src/features. Backend services under backend/app/services are grouped by document processing, retrieval, providers, connectors, operations, security, storage, and application flows. The reorganization preserves API contracts and requirement IDs.

For local development, activate the virtual environment from backend and run uvicorn app.main:app --host 0.0.0.0 --port 8000. In another terminal, run npm run dev -- --host 0.0.0.0 --port 5173 --strictPort from frontend/app. Run the document worker with python -m app.workers.document_worker and the connector scheduler with python -m app.workers.connector_scheduler_worker in separate processes.

The upload queue shows each file’s transfer state. A finished transfer does not imply that processing is complete. Uploads, set saves, and conversation deletions have no global success notification. Copy and feedback confirmations appear briefly next to their controls. Operation errors must still be shown.

3.1 External Dependencies

The language model and OCR provider are configurable external dependencies. Disabling OCR affects only paths that require it. A model outage must leave sign-in, document browsing, and saved conversations available. In a multi-instance deployment, all instances need shared access to the database and storage.

3.2 External Data Policy

A file or text excerpt may be sent to an external service for processing. Enabling a provider depends on organizational data policy. A provider name in the interface does not establish that credentials, quota, or an active connection exist.

Chapter 4

Roles and Business Rules

4.1 Roles and Access Matrix

Capability	Administrator	User	Operations
Chat and view sources	Yes (within the organization)	Yes (within authorized scope)	No
Manage knowledge base and documents	Yes (within the organization)	No	No
Manage members	Yes	No	With resource permission
Use an active assistant	Yes	Yes	No
Manage assistants and connectors	Yes	No	Yes
Change own password	Yes	Yes	Yes
Organization management reports	Yes	No	No

4.2 Business Rules

BR-01 Organization Boundary

The system shall derive the user’s organization from a valid session and shall disregard any organization identifier supplied by the client as proof of access.

Acceptance criteria: Changing an ID in a URL or request body cannot give a user in organization A access to documents, conversations, members, or notifications in organization B.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-01

BR-02 All Sources Scope

The All option shall include the union of the user’s authorized sets and exclude all other sets from retrieval.

Acceptance criteria: When two of three sets are authorized, an All query returns no chunk from the third.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-02

BR-03 Administration Interface Restrictions

Knowledge base and team management options shall be hidden from ordinary users. Direct routes shall also enforce permissions.

Acceptance criteria: Opening an administrative URL as a regular user redirects to an authorized page or shows an access error. Direct API calls are denied.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-03

BR-04 Deletion Confirmation

A deletion prompt shall name the target and explain the effect. Deletion shall run only after confirmation.

Acceptance criteria: Cancelling deletion of a set, conversation, or member results in no effective deletion request.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-04

BR-05 Access After a Permission Change

Each new request shall apply current permissions. Cached data shall not perpetuate previous access.

Acceptance criteria: After a set permission is revoked, subsequent source requests are denied or return no result from that set.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-05

BR-06 Administrator Account Protection

The regular member deletion flow shall not delete administrator accounts.

Acceptance criteria: No delete button appears for an administrator, and direct deletion requests are denied.

Priority: Essential | Status: Acceptance baseline | Reference: C01, U | Test: AT-BR-06

Chapter 5

Authentication and Session

FR-AUTH-01 Valid Sign-in

The system shall validate the username and password and create a valid session only upon success.

Acceptance criteria: Correct credentials open the workspace. An incorrect password creates no usable session.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-01

FR-AUTH-02 Sign-in Errors

Sign-in errors shall not disclose account existence or internal details.

Acceptance criteria: An unknown username and an incorrect password reveal no additional information about whether an account exists.

Priority: Essential | Status: Acceptance baseline | Reference: U | Test: AT-FR-AUTH-02

FR-AUTH-03 Failed Attempt Limits

The system shall rate-limit failed sign-in attempts by account and request address.

Acceptance criteria: Exceeding the configured threshold temporarily limits attempts. Other accounts can continue normal sign-in in line with policy.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-03

FR-AUTH-04 Session Expiry

Sessions shall have a limited lifetime. Remember Me shall follow the configured duration.

Acceptance criteria: The API rejects expired sessions, and the interface redirects users to sign-in.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-04

FR-AUTH-05 Secure Session Storage

The session identifier shall reside in an HttpOnly cookie with security attributes appropriate to the environment.

Acceptance criteria: Page scripts cannot read the session cookie. Secure is enabled in production.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-05

FR-AUTH-06 Confirmed Sign-out

Sign-out shall prompt for confirmation and revoke the server-side session once confirmed.

Acceptance criteria: Cancelling keeps the user signed in. The previous session is rejected after sign-out.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-06

FR-AUTH-07 Change Password

Users shall be able to change their password in settings using the valid current password and matching confirmation of the new password.

Acceptance criteria: An invalid current password or mismatched confirmation is rejected. A valid new password works, and other sessions are revoked according to policy.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-07

FR-AUTH-08 Keep Credentials Out of Logs

The system shall not record raw passwords, cookies, or tokens in logs, notifications, errors, or analytics exports.

Acceptance criteria: Reviewing successful and failed sign-in and password changes exposes no raw credentials.

Priority: Essential | Status: Acceptance baseline | Reference: C02, U | Test: AT-FR-AUTH-08

Chapter 6

Organization Member Management

FR-USER-01 Create Member

An administrator shall be able to create a new member with valid fields within their organization.

Acceptance criteria: Duplicate names and invalid input are rejected. The new member belongs to the administrator’s organization.

Priority: Essential | Status: Acceptance baseline | Reference: C03, U | Test: AT-FR-USER-01

FR-USER-02 List and Search

The member list shall support search and role filtering.

Acceptance criteria: Combined search and role filters return only matching members of the same organization, with a clear empty state.

Priority: Essential | Status: Acceptance baseline | Reference: C03, U | Test: AT-FR-USER-02

FR-USER-03 Member Ordering

Administrators shall appear first, followed by users in ascending creation order.

Acceptance criteria: Refresh and filtering preserve ordering. Equal timestamps are ordered by ID for stability.

Priority: Essential | Status: Acceptance baseline | Reference: C03, U | Test: AT-FR-USER-03

FR-USER-04 Delete Member

Regular members shall be deleted only after confirmation, and their sessions shall no longer be usable.

Acceptance criteria: The deleted member no longer appears in the list, and requests using their old session are rejected. Treatment of earlier conversations is recorded in accordance with retention policy.

Priority: Essential | Status: Acceptance baseline | Reference: C03, U | Test: AT-FR-USER-04

FR-USER-05 Deletion Control

The delete action shall use a trash icon with an accessible name and an adequate touch target.

Acceptance criteria: The screen reader name identifies the delete action and member. The touch target is at least 44 by 44 pixels.

Priority: Essential | Status: Acceptance baseline | Reference: C03, U | Test: AT-FR-USER-05

Chapter 7

Knowledge Sets and Resource Permissions

FR-KB-01 Create a Set

Administrators shall be able to create a set with a valid name and optional description.

Acceptance criteria: Blank or whitespace-only names are rejected. A valid new set appears with zero documents.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-01

FR-KB-02 Edit a Set

Editing the name and description shall preserve the set ID and document associations.

Acceptance criteria: After a rename, documents and permissions remain associated, and selectors show the new name.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-02

FR-KB-03 Set Permissions

The API and retrieval service shall enforce view, edit, and manage permission levels.

Acceptance criteria: A viewer cannot add documents or change another user’s permission.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-03

FR-KB-04 Select a Set

The selected set shall define the upload target, document list, and Knowledge Assistant query scope.

Acceptance criteria: Changing the set updates the list and assistant title. A file is not added to the previously selected set.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-04

FR-KB-05 Delete a Set

Each set card shall provide a delete control and explain the effect before confirmation.

Acceptance criteria: Cancellation changes nothing. Confirmation removes the set from selectors and prevents further searches of that set.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-05

FR-KB-06 Document Counts

Total and ready document counts shall be computed from stored states.

Acceptance criteria: Queued and failed documents are excluded from the ready count. The count updates after indexing.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-06

FR-KB-07 Search and Status

The document list shall support name, date, status, search, and status filtering.

Acceptance criteria: The failed-document filter returns the correct results. Clearing search restores the complete authorized list.

Priority: Essential | Status: Acceptance baseline | Reference: C04, U | Test: AT-FR-KB-07

Chapter 8

Secure Upload and File Quarantine

The current code configures a per-file limit of 10 × 1024 × 1024 bytes. Content validation covers PDF, JPEG, PNG, TIFF, and UTF-8 text. Release tests shall verify agreement between the UI and API allowlists. The UI label “10 MB” shall correspond to the same byte limit.

FR-UP-01 File Transfer

An authorized user shall be able to upload files to the selected set and see each file’s transfer status.

Acceptance criteria: During a multi-file upload, success for one file does not conceal an error for another.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-01

FR-UP-02 Size Limit

The API shall enforce size while receiving data and shall not rely solely on the client or Content-Length.

Acceptance criteria: Empty files and files one byte over the limit are rejected. Valid content at the limit is accepted.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-02

FR-UP-03 Content Type Matching

The extension, MIME type, and file signature shall agree with a supported type.

Acceptance criteria: An executable with a PDF extension or an image with an incompatible MIME type does not enter processing and produces a file type error.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-03

FR-UP-04 File Name and Path

The original name shall be retained for display, while the storage path shall be derived from a safe identifier.

Acceptance criteria: Relative paths, path separators, and Persian characters in names cannot create files outside the storage area.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-04

FR-UP-05 Quarantine Before Processing

Incoming files shall be quarantined before processing and moved only after validation and a clean scan.

Acceptance criteria: No extraction or OCR job runs during scanning. The file cannot be retrieved from a public path.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-05

FR-UP-06 Production Malware Scan

Scanning shall be mandatory in production, and malware detection shall stop document acceptance.

Acceptance criteria: A controlled EICAR test in the test environment rejects the file and creates no chunk in the vector store.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-06

FR-UP-07 Scanner Failure

If scanning is required, a missing or invalid scanner response shall temporarily reject the file.

Acceptance criteria: A ClamAV outage returns an actionable response. The file does not enter the queue on the assumption that it is clean.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-07

FR-UP-08 Record Rejection Reason

The system shall record the rejection reason, user and organization IDs, type, size, and fingerprint without logging raw content.

Acceptance criteria: Invalid type, malware detection, and scanner unavailability are distinguishable. Logs contain no file content or secrets.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-08

FR-UP-09 Infected File Retention

Retention or cleanup of detected files shall follow configuration and restricted quarantine access.

Acceptance criteria: With retention disabled, the file is removed. If retained, it cannot be publicly downloaded or processed.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-09

FR-UP-10 Separate Transfer from Processing

Finishing a transfer shall not mark the document ready for answering.

Acceptance criteria: After transfer, the interface shows queued or processing status. Ready appears only after indexing finishes.

Priority: Essential | Status: Acceptance baseline | Reference: C05, U | Test: AT-FR-UP-10

Chapter 9

Document Processing and OCR

FR-ING-01 Durable Job Registration

Accepting a document shall create a trackable job in a durable queue.

Acceptance criteria: After restarting the API, the accepted document and its job status are recoverable.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-01

FR-ING-02 Atomic Job Claim

A worker shall claim a job atomically so two workers cannot execute it concurrently.

Acceptance criteria: Starting two workers against the same job results in only one valid owner.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-02

FR-ING-03 Independent Heartbeat

Long-running processing shall not interrupt job and worker heartbeats.

Acceptance criteria: When OCR takes longer than the heartbeat interval, the worker remains current and the job is not mistakenly reclaimed.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-03

FR-ING-04 Text Extraction

Text-based PDFs and valid TXT files shall first use the supported text extraction path.

Acceptance criteria: A sample text document produces nonempty, comparable text and is not sent to OCR unnecessarily.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-04

FR-ING-05 OCR Selection

A document requiring OCR shall use only the active provider with the required configuration.

Acceptance criteria: Disabled mode records a clear message. When a provider is selected, the file is not sent to another provider.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-05

FR-ING-06 OCR Limits

Page count, timeout, and the result polling interval shall be bounded and configurable.

Acceptance criteria: A document above the page limit or an OCR timeout produces an actionable state. Polling does not continue indefinitely.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-06

FR-ING-07 Extraction Failure

Empty output, a corrupt file, or OCR failure shall not mark a document ready.

Acceptance criteria: A failed document has an understandable reason and does not appear in ready-document search results.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-07

FR-ING-08 Stage Progress

Reading, OCR, chunking, and indexing status shall reflect the actual processing stage.

Acceptance criteria: If a stage stalls, its time and cause are traceable. Artificial percentages do not replace actual results.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-08

FR-ING-09 Bounded Retries

Transient errors shall have bounded retries with increasing delays. Permanent errors shall not cause endless retries.

Acceptance criteria: A temporary network outage is handled within the retry limit. An invalid file is not retried indefinitely.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-09

FR-ING-10 Manual Reprocessing

An administrator shall be able to reprocess a failed document after resolving the cause.

Acceptance criteria: Two simultaneous retry clicks do not create duplicate effective jobs. The new attempt’s result is distinguishable.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-10

FR-ING-11 Index Integrity

The indexed state shall be set only after all chunks and their links to the document are stored.

Acceptance criteria: A partial upsert failure does not mark the document ready. Retrieval returns only a complete, authorized version.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-11

FR-ING-12 Deletion During Processing

At job completion, the system shall recheck whether the document was deleted or detached from its set.

Acceptance criteria: After deletion, the worker does not restore the document to the list or create unintended accessible chunks.

Priority: Essential | Status: Acceptance baseline | Reference: C06, U | Test: AT-FR-ING-12

Chapter 10

Chunking and Document Content Review

FR-CHK-01 Profile Selection

Administrators shall be able to select the precise, balanced, or broad-context profile for subsequent processing.

Acceptance criteria: The profile change is saved. Existing documents are clearly identified as requiring reprocessing to apply the new setting.

Priority: Essential | Status: Acceptance baseline | Reference: C07, U | Test: AT-FR-CHK-01

FR-CHK-02 Parameter Validation

Chunk size, overlap, and parent size shall satisfy contract limits. Overlap shall be smaller than the chunk.

Acceptance criteria: The API rejects overlap equal to or larger than the child size, even when the client is bypassed.

Priority: Essential | Status: Acceptance baseline | Reference: C07, U | Test: AT-FR-CHK-02

FR-CHK-03 Chunk Traceability

Every chunk shall link to its document, processing version, authorized scope, and page or parent when available.

Acceptance criteria: Opening an answer source displays the correct document. Unattributed chunks cannot be retrieved.

Priority: Essential | Status: Acceptance baseline | Reference: C07, U | Test: AT-FR-CHK-03

FR-CHK-04 Document Inspector

Authorized administrators shall be able to view and search parent blocks and child chunks.

Acceptance criteria: Persian text, filenames, and the parent-child structure remain legible. Unauthorized users cannot read them.

Priority: Essential | Status: Acceptance baseline | Reference: C07, U | Test: AT-FR-CHK-04

FR-CHK-05 Chunking Method Changes

Changes in hierarchical or semantic chunking shall be identifiable by processing version.

Acceptance criteria: Model version and method are recorded. An answer does not inadvertently mix old and new chunks.

Priority: Essential | Status: Acceptance baseline | Reference: C07, U | Test: AT-FR-CHK-05

Chapter 11

Retrieval and Answer Generation

FR-RAG-01 Question Scope

For each question, the server shall determine the scope of sets, documents, and the selected assistant where applicable.

Acceptance criteria: Changing a document ID in the request cannot expand authorized scope.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-01

FR-RAG-02 Pre-Retrieval Filtering

Organization and authorized-document filters shall be applied during retrieval before model context is built.

Acceptance criteria: An unauthorized chunk is never sent to the model, even if it ranks first in similarity.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-02

FR-RAG-03 Metadata Filtering

Metadata filters and the selection of all or specific documents shall affect retrieval.

Acceptance criteria: Identical questions with opposing filters cite only documents that match their respective filters.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-03

FR-RAG-04 Sources-Only Answers

In sources mode, the answer shall stay within available evidence and explicitly acknowledge missing information.

Acceptance criteria: A question unanswered by the set is not filled in with invented facts or irrelevant citations.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-04

FR-RAG-05 Hybrid Answers

In hybrid mode, model knowledge shall be distinguishable from evidence-backed claims.

Acceptance criteria: The interface labels hybrid answers appropriately and does not fabricate sources for unsupported portions.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-05

FR-RAG-06 Grounding Status

Answers shall communicate grounding status clearly. Source count alone does not guarantee correctness.

Acceptance criteria: An unsupported answer is not presented as definitively grounded because it has an unrelated chunk.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-06

FR-RAG-07 Source Citation

A citation shall include the document name, relevant chunk, and page number when available.

Acceptance criteria: Clicking the source opens the supporting text. A missing page number is not invented.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-07

FR-RAG-08 Removed Source

If a source is later deleted or its permission revoked, subsequent views shall apply current permissions.

Acceptance criteria: Confidential text is not returned from the source cache. The interface explains unavailability without exposing content.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-08

FR-RAG-09 Conversation Context

Follow-up questions shall use authorized history from the same conversation without leaking other conversations.

Acceptance criteria: “What is their role?” is interpreted using the current conversation. A name from another user’s conversation is not introduced.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-09

FR-RAG-10 Model Error Handling

Quota errors, timeouts, and invalid model responses shall produce an actionable failure.

Acceptance criteria: The interface exits the waiting state and does not present an empty response as success or substitute invented text.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-10

FR-RAG-11 Instructions Inside Documents

Retrieved text shall be treated as data and shall not alter system instructions or access boundaries.

Acceptance criteria: A test document asking to reveal secrets or bypass permissions does not cause disclosure or an unauthorized call.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-11

FR-RAG-12 Answer Mode and Budget

Quick Answer and Deep Research shall have defined scopes and processing budgets.

Acceptance criteria: Switching modes changes the answer path as specified. Deep Research does not become an unbounded chain of model calls.

Priority: Essential | Status: Acceptance baseline | Reference: C08, U | Test: AT-FR-RAG-12

Chapter 12

Conversation and Answer Control

FR-CHAT-01 New Conversation

Users shall be able to start a conversation without overwriting earlier history.

Acceptance criteria: A new conversation gets its own ID and does not inherit messages from previous conversations.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-01

FR-CHAT-02 Save History

User messages, answers, sources, and timestamps shall be stored with the conversation owner and organization.

Acceptance criteria: Reopening the conversation shows the same order of messages and recorded sources.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-02

FR-CHAT-03 Single Submission

The interface shall prevent accidental repeat submissions while a question is active.

Acceptance criteria: A rapid double click or repeated Enter does not generate multiple unwanted answers for one action.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-03

FR-CHAT-04 Waiting State

While an answer is pending, the interface shall show a concise, legible status and a way to cancel.

Acceptance criteria: Send becomes a stop control whose action is announced by screen readers.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-04

FR-CHAT-05 Cancel Generation

Cancellation shall stop the waiting state and request connection. Late results from cancelled requests shall not be shown.

Acceptance criteria: The interface is usable after stopping, and a late answer does not appear in a new conversation. External processing already underway may still incur costs.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-05

FR-CHAT-06 Preserve Input on Error

After server failure or cancellation, question text shall remain available for editing or retry.

Acceptance criteria: A network outage does not discard typed text or leave the composer permanently disabled.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-06

FR-CHAT-07 Rename and Delete

Users shall be able to rename their conversations or delete them after confirmation.

Acceptance criteria: Cancelling deletion preserves data. The renamed title matches in the recent list and conversation view.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-07

FR-CHAT-08 Copy Answer

A short message shall appear below the copy icon only after the clipboard operation succeeds. No global success message shall appear.

Acceptance criteria: Clipboard permission denial produces no success message. Users can still read or select the text.

Priority: Essential | Status: Acceptance baseline | Reference: C09, U | Test: AT-FR-CHAT-08

Chapter 13

Custom Assistants

FR-AST-01 Manage Assistants

Administrators shall be able to create or edit an assistant’s name, description, instructions, model, answer mode, and sources.

Acceptance criteria: Invalid required fields are not saved. Reopening the form shows the saved values.

Priority: Essential | Status: Acceptance baseline | Reference: C10, U | Test: AT-FR-AST-01

FR-AST-02 Assistant Scope

Assistant sources shall be intersected with the current user’s permissions.

Acceptance criteria: Linking a confidential set to an assistant does not make it searchable to a user without permission.

Priority: Essential | Status: Acceptance baseline | Reference: C10, U | Test: AT-FR-AST-02

FR-AST-03 Enable and Disable

A disabled assistant shall be unavailable for new conversations.

Acceptance criteria: A direct request to start a conversation with a disabled assistant is also denied. Prior history remains available in line with policy.

Priority: Essential | Status: Acceptance baseline | Reference: C10, U | Test: AT-FR-AST-03

FR-AST-04 Unavailable Model

An unavailable model list or missing credentials shall produce a clear error state.

Acceptance criteria: Model loading errors do not silently save invalid settings or display fictional models.

Priority: Essential | Status: Acceptance baseline | Reference: C10, U | Test: AT-FR-AST-04

FR-AST-05 Record Conversation Association

Conversations started with an assistant shall retain its ID and effective settings for traceability.

Acceptance criteria: Feedback and analytics can attribute the answer to its assistant. Later changes do not obscure the historical settings.

Priority: Essential | Status: Acceptance baseline | Reference: C10, U | Test: AT-FR-AST-05

Chapter 14

Connectors and Synchronization

FR-CON-01 Define a Connection

Administrators shall be able to attach a supported connector to a set with type, address, and sync schedule.

Acceptance criteria: Incomplete settings are rejected. The connector type shown matches actual API capability.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-01

FR-CON-02 Provider Prerequisites

SharePoint, Google Drive, S3, web, and GitHub connectors shall check required settings and permissions before sync.

Acceptance criteria: Missing credentials or permissions produce an actionable error without creating random data or false success.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-02

FR-CON-03 Safe URL Fetching

URL-based connectors shall enforce address and network policy and block unintended access to internal resources.

Acceptance criteria: Unauthorized destinations such as metadata endpoints, loopback, or prohibited private addresses are rejected. Redirects are also checked.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-03

FR-CON-04 Scheduling

The scheduler shall run due connectors on their enabled schedules.

Acceptance criteria: Disabled and not-yet-due connectors do not run. Stopping the scheduler does not erase the recorded execution time.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-04

FR-CON-05 Synchronization Lease

Lease acquisition and renewal shall be atomic and independent of long network operations.

Acceptance criteria: Two schedulers cannot execute the same connector concurrently. The lease can be recovered after its owner stops.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-05

FR-CON-06 Incremental Sync

An unchanged source shall not create a duplicate document on every sync.

Acceptance criteria: Two consecutive syncs of the same source do not increase document count. A real change produces a traceable new version.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-06

FR-CON-07 External Source Deletion

Deletion at the source shall follow the connector contract and be recorded in sync status.

Acceptance criteria: An authorized delete event removes the document from retrieval according to policy without affecting other connectors’ documents.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-07

FR-CON-08 Webhook

Webhook delete and upsert events shall verify connector identity, organization, and message authenticity.

Acceptance criteria: Invalid messages and messages belonging to another connector are rejected. Replays have no unintended duplicate effect.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-08

FR-CON-09 Partial Failure

A source or document error shall be recorded independently of other results.

Acceptance criteria: Losing access to one file does not report the entire sync as successful or remove healthy documents without cause.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-09

FR-CON-10 Connector Secrets

Secrets shall be retrieved from secure storage when needed and never returned by forms or API responses.

Acceptance criteria: Opening a form, checking logs, and running a sync expose no raw token. Secret requests use the correct organization.

Priority: Essential | Status: Acceptance baseline | Reference: C11, U | Test: AT-FR-CON-10

Chapter 15

Conversation Sharing and Feedback

FR-SHR-01 Create a Share

Authorized users shall be able to create a read-only conversation snapshot for the team or via a link.

Acceptance criteria: Recipients cannot change messages or access data outside the snapshot.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-01

FR-SHR-02 Team Scope

Team sharing shall verify the session and authorized membership.

Acceptance criteria: A user outside the organization cannot read the conversation with a team token.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-02

FR-SHR-03 Public Link

Creating a link that works without a session shall explain the implications of publishing the snapshot.

Acceptance criteria: The creator sees the publication scope first. The snapshot contains no credentials or internal metadata.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-03

FR-SHR-04 Revoke a Share

The authorized owner shall be able to revoke the link. The server shall then stop serving its content.

Acceptance criteria: A revoked link does not work in an anonymous session, and its content cannot be recovered from the application response cache.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-04

FR-SHR-05 Snapshot Stability

Later messages shall not enter a published snapshot without a new action.

Acceptance criteria: After a new message is added, the old link still displays the shared version.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-05

FR-SHR-06 Submit Feedback

Users shall be able to register positive or negative feedback for a specific answer and see the actual result in a short message next to the control. No global success message shall appear.

Acceptance criteria: An API error produces no success message. Feedback is linked to the correct message, conversation, and organization.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-06

FR-SHR-07 Avoid Double Counting

Submitting or changing feedback again shall not accidentally count twice in satisfaction metrics.

Acceptance criteria: Repeated feedback from the same user for the same answer is handled by the uniqueness or replacement contract.

Priority: Essential | Status: Acceptance baseline | Reference: C12, U | Test: AT-FR-SHR-07

Chapter 16

Analytics and Report Output

This chapter covers product-level analytics and reporting requirements. The version-specific API contract defines exact metrics and calculations. Acceptance criteria shall cover data categories, access scope, date ranges, and export behavior.

> Note: All analytics metrics shall have negative tests for organization boundaries. Changing identifiers cannot expose another organization’s analytics.

Chapter 17

Product Notifications and Operational Alerts

FR-NOT-01 Document Completion Event

Document readiness or processing failure shall create a persistent notification for the relevant user.

Acceptance criteria: The user can return later and see the result and related target. Other users do not see the notification.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-01

FR-NOT-02 Sync Completion Event

Sync success or failure shall provide an understandable result and authorized link to the source.

Acceptance criteria: Connector failure is distinguishable from document processing failure. Opening the target rechecks authorization.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-02

FR-NOT-03 Mark Notifications Read

Marking one or all notifications read shall update the interface only after server confirmation.

Acceptance criteria: A network failure does not permanently mark a notification read. Repeated requests do not corrupt counts.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-03

FR-NOT-04 Bell Placement

The notification bell shall appear only in the workspace, aligned with the reporting-period control in the administration view.

Acceptance criteria: There is no global bell in chat, documents, members, or settings. The workspace panel stays inside the viewport.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-04

FR-NOT-05 Panel Appearance

The notification panel shall offer sufficient contrast in light and dark themes and an accessible close control.

Acceptance criteria: The light theme uses a light background and legible text. Escape and outside clicks close the panel.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-05

FR-NOT-06 Email Alerts

Important operational errors shall send nonblocking alerts when SMTP and recipients are configured.

Acceptance criteria: SMTP downtime does not hold up the original request. Delivery failure is identifiable in logs.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-06

FR-NOT-07 Alert Deduplication

Repeated errors shall have a defined cooldown to prevent an email flood.

Acceptance criteria: Multiple events with one key in a short interval do not cause unlimited messages. Cross-process deduplication is tested separately.

Priority: Essential | Status: Acceptance baseline | Reference: C14, U | Test: AT-FR-NOT-07

Chapter 18

User Interface and Localization

NFR-UI-01 Two Languages

Persian and English shall control interface text, page direction, and displayed dates.

Acceptance criteria: Switching language does not mix inappropriate labels. Document dates use the Solar Hijri calendar in Persian.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-01

NFR-UI-02 Bilingual Filenames

Filename alignment shall follow the interface language, while the filename’s text direction shall follow its content.

Acceptance criteria: A Persian filename with a PDF extension appears on the appropriate side of its icon in English. The extension and name remain correctly ordered in Persian.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-02

NFR-UI-03 Typography

Functional text shall be at least 12 px and mobile input text at least 16 px.

Acceptance criteria: Long names, Persian text, and multiline questions display without clipping essential content.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-03

NFR-UI-04 Theme Preview

The light preview shall show a light appearance, and the dark preview a dark appearance, regardless of the active theme.

Acceptance criteria: In dark settings, the light preview still has a recognizable light background.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-04

NFR-UI-05 Loading and Errors

Every long request shall have an appropriate waiting state, error state, and recovery option.

Acceptance criteria: API downtime does not produce endless waiting or a blank page. Retries depend on user choice or a bounded policy.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-05

NFR-UI-06 Persist Preferences

Language and theme shall persist after reload and apply to dates and popups.

Acceptance criteria: Reloading a direct route restores saved language and theme without leaving an inappropriate color state.

Priority: Essential | Status: Acceptance baseline | Reference: C15, U | Test: AT-NFR-UI-06

Chapter 19

Responsiveness and Accessibility

19.1 Width Ranges and Baseline Behavior

Range	CSS width (px)	Baseline behavior
Small mobile	320-479	Single column and touch controls
Large mobile	480-677	Drawer and full-width panel
Tablet	768-1023	Layout adapts to available space
Laptop	1024-1439	Multiple columns when panels fit
Wide desktop	1440 and above	Readable width and controlled spacing

NFR-A11Y-01 Main Flow Coverage

Sign-in, conversations, documents, upload, sidebar, settings, and members shall be usable at reference widths.

Acceptance criteria: Essential controls stay within the viewport at 320, 375, 390, 768, 1024, and 1440 px.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-01

NFR-A11Y-02 Mobile Sidebar

At narrow widths, the sidebar shall be a dismissible drawer. Content behind it shall be noninteractive while open.

Acceptance criteria: Tab stays inside the drawer. Escape closes it, and focus returns to the opening control.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-02

NFR-A11Y-03 Mobile Knowledge Panel

At narrow widths, the Knowledge Assistant shall use a full-width panel or bottom sheet.

Acceptance criteria: Opening the mobile keyboard does not hide the Send or Stop controls.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-03

NFR-A11Y-04 Tables and Toolbars

Tables shall use contained horizontal scrolling or a card view. Toolbars shall be able to wrap.

Acceptance criteria: Narrow layouts avoid whole-page horizontal scrolling and retain a clear control order.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-04

NFR-A11Y-05 Keyboard and Focus

All primary operations shall work with a keyboard and visible focus.

Acceptance criteria: Users can sign in, type a question, open a source, and close a dialog without a mouse. Focus does not move behind a modal.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-05

NFR-A11Y-06 Control Names and Touch Targets

Icon-only controls shall have accessible names and touch targets of at least 44 by 44 px.

Acceptance criteria: A screen reader announces each icon’s action. Touch target size is measured independently of icon size.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-06

NFR-A11Y-07 Contrast

The interface shall target WCAG 2.2 Level AA for text and essential components.

Acceptance criteria: Normal text reaches at least 4.5:1. Large text and necessary components reach at least 3:1. Results are recorded for each theme.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-07

NFR-A11Y-08 Reduced Motion

The prefers-reduced-motion setting shall remove or reduce nonessential motion.

Acceptance criteria: Answer waiting states and page changes remain recognizable without strong animation.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-08

NFR-A11Y-09 Status Announcements

Loading, errors, and operation outcomes shall be understandable to screen reader users.

Acceptance criteria: Upload completion or submission failure is announced without forcibly moving focus or producing disruptive repeated messages.

Priority: Essential | Status: Acceptance baseline | Reference: R02, U | Test: AT-NFR-A11Y-09

Chapter 20

Data Contract and Lifecycle

20.1 Entity Map

Entity	Required data and relationships
Organization and user	Stable ID, role, state, creation time, and organization membership
Session	Owner, expiration, revocable ID, and revocation time
Knowledge set	Name, description, organization, and resource permissions
Document	Original name, type, size, safe path, status, and processing version
Processing job	Document, requester, attempts, lease, and heartbeat
Chunk	Document and version, text, parent, page if available, and vector
Conversation and message	Owner, order, text, time, sources, and effective settings
Assistant	Instructions, model, answer mode, status, and associated sets
Connector and sync	Type, target, secret reference, schedule, lease, and result
Feedback and share	Related answer or snapshot, owner, scope, and revocation
Notification	User, organization, event, severity, target, and read time

DAT-01 Stable Identifier

Each operational entity shall have a stable ID independent of its display name.

Acceptance criteria: Renaming a file, set, or conversation does not break data relationships.

Priority: Essential | Status: Acceptance baseline | Reference: C01..C17 | Test: AT-DAT-01

DAT-02 Data Consistency

Analytics, answer sources, and conversation messages shall use a consistent source with stable identity.

Acceptance criteria: A document count or ownership mismatch across indexes produces a detectable error instead of contradictory reports.

Priority: Essential | Status: Acceptance baseline | Reference: C13, C16 | Test: AT-DAT-02

DAT-03 Processing Versioning

Each document version shall retain the effective processing version, model, and chunking method.

Acceptance criteria: After reprocessing, older answers are not incorrectly overwritten with new chunks. Any loss of access is explicit.

Priority: Essential | Status: Acceptance baseline | Reference: C06 | Test: AT-DAT-03

DAT-04 Reprocessing

Reprocessing shall affect only the new version. The previous version shall remain recoverable according to policy.

Acceptance criteria: An older answer is not overwritten with a new chunk after reprocessing.

Priority: Essential | Status: Acceptance baseline | Reference: C06 | Test: AT-DAT-04

DAT-05 Coordinated Cleanup

Deletion shall define what happens to associated records, files, vectors, shares, and notifications.

Acceptance criteria: After final deletion, searches and old links do not return deleted content. Cleanup failures can be retried.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C06 | Test: AT-DAT-05

DAT-06 Data Retention

Retention periods for documents, conversations, events, and backups shall be defined per organization and policy.

Acceptance criteria: Cleanup affects only data outside the approved period. Chapter 27 decisions are resolved before production.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C06 | Test: AT-DAT-06

20.2 Document Lifecycle

The lifecycle begins with receipt and quarantine. Accepted documents are queued, then undergo extraction, OCR when needed, chunking, and indexing. The successful end state is indexed. Security rejection before acceptance and failed processing after acceptance are distinct states and require distinct messages.

Conceptual state	Event	Expected behavior
Quarantine	Clean approval or rejection	No extraction before approval
Queued	Worker claim	Waiting is traceable by job ID
Processing	Completion or error	Update stage and maintain heartbeat
Ready	Deletion or reprocessing	Use only the valid version in retrieval
Failed	Retry or deletion	Preserve reason and attempt count
Cancelled or deleted	Cleanup	Late results do not become accessible again

Chapter 21

Interface Contracts and Errors

OpenAPI defines the versioned REST API contract. Detailed field schemas shall come from that version of the API. This document fixes interface families and behavioral semantics. Errors shall have a classifiable code and safe message and shall be correlatable to a request ID for operational tracking.

21.1 Interface Families

Interface family	Responsibility	Error contract
users and auth	Sign-in, sign-out, password change, and members	Unauthenticated, forbidden, or invalid input
documents and document sets	Sets, files, status, and retry	Type, size, malware, or unavailable scanner
rag, research, and conversations	Conversation, retrieval, and answer	Unauthorized source, timeout, or model failure
connectors	Registration, webhook, and sync	Credentials, blocked destination, or partial failure
feedback and chat shares	Sharing and feedback	Invalid data, revocation, or unauthorized owner
notifications and analytics	Reporting and notifications	Unauthorized organization or user
metrics, ready, and health	Health and observability	Not ready or metrics authorization

21.2 Common Error Codes

Common code	Meaning and interface response
400 or 422	Input or file rejected. Show a specific message and allow correction.
401	Invalid session. Require sign-in without disclosing data.
403 or 404	Unauthorized or missing source under the nondisclosure policy.
413	Size exceeds the limit. Show the permitted limit.
415	Unsupported or mismatched content type.
429	Rate limited. Respect the reported retry time.

503	A required dependency is temporarily unavailable. Do not report success.
Network error or timeout	Preserve input, end waiting, and allow a controlled retry.

> Note: Bound connection and response timeouts for external calls. Retry only transient errors when the operation can be repeated safely. Automatically retrying job creation or link publication without a unique identifier can duplicate effects. The API contract shall address this.

Chapter 22

Security and Privacy

SEC-01 Authorization at Every Layer

Authorization shall be enforced in the API, services, retrieval, and file downloads.

Acceptance criteria: Bypassing the interface or changing an identifier does not create an alternative disclosure path.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-01

SEC-02 Transport Protection

Production connections shall use valid TLS. Insecure settings shall be restricted to testing.

Acceptance criteria: Production configuration rejects insecure cookies or disabled scanning.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-02

SEC-03 Browser Protection

CORS, mutating requests, and Markdown content shall be controlled to prevent unintended requests and code execution.

Acceptance criteria: Unauthorized origins cannot perform sensitive operations with a user’s session. Model or document text cannot execute scripts.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-03

SEC-04 Minimize External Data

Model and OCR requests shall send only necessary data to approved providers.

Acceptance criteria: An outbound request test finds no out-of-scope documents or service secrets in the payload.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-04

SEC-05 Protect Secrets

Secrets shall be kept out of public settings, source control, and error responses and shall support rotation.

Acceptance criteria: Keys can be rotated without rewriting user documents. Logs reveal neither the old nor new values.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-05

SEC-06 Audit Sensitive Operations

Failed sign-in, password change, permission change, deletion, and security rejection of files shall be traceable.

Acceptance criteria: Events include time, actor, organization, target, and outcome. Passwords and complete document text are not recorded.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-06

SEC-07 Consumption Limits

Uploads, sign-in, questions, and external connections shall have suitable rate limits or budgets.

Acceptance criteria: High usage cannot trigger an unbounded request loop. Limit errors are distinguishable from internal failures.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-07

SEC-08 Data Boundary Testing

A release shall pass negative tests with at least two organizations and multiple permission levels.

Acceptance criteria: Document, source, conversation, share, notification, and member paths are tested with another organization’s identifiers. Outcomes are recorded.

Priority: Essential | Status: Acceptance baseline | Reference: C01, C05, U | Test: AT-SEC-08

Chapter 23

Reliability, Monitoring and Recovery

OPS-01 Process Liveness

The liveness check shall confirm the process is running without creating or deleting data.

Acceptance criteria: Repeated health calls do not change collection, document, or file counts.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-01

OPS-02 Service Readiness

Readiness shall check essential dependencies and worker freshness with bounded timeouts.

Acceptance criteria: PostgreSQL or Qdrant outages, or expired heartbeats, return 503. A live process may still pass its health check.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-02

OPS-03 Operational Metrics

Processing duration, queue length, OCR and model errors, API usage, and sync failures shall be observable.

Acceptance criteria: A sample event changes the relevant counter or measurement. Labels contain neither secrets nor unbounded values.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-03

OPS-04 Metrics Access

Internal metrics shall require explicit authorization and shall not be public when security settings are absent.

Acceptance criteria: Unauthenticated requests cannot access detailed metrics.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-04

OPS-05 Centralized Logging

Services shall produce collectable logs with request identifiers.

Acceptance criteria: A document failure can be traced from API to worker. Local logs alone do not constitute a central logging deployment.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-05

OPS-06 Coordinated Backup

Backups shall cover PostgreSQL, files, and the data needed to rebuild vectors and settings.

Acceptance criteria: An isolated restore reconnects documents, permissions, and answer sources. Secrets are protected separately.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-06

OPS-07 Recovery Test

Recovery shall be rehearsed with a real backup in an isolated environment. Record elapsed time and data loss.

Acceptance criteria: Backup success alone does not pass the test. Verify sign-in, document reading, and source-grounded answers after restore.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-07

OPS-08 Version Upgrade

Data migration and deployment shall have documented prerequisites, execution order, and a rollback path.

Acceptance criteria: A migration failure does not route traffic to an incompatible version. Rollback with data recovery can be rehearsed.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-08

OPS-09 Resource Usage Control

Polling, the scheduler, and workers shall have concurrency controls. Builds and heavy tests shall stay outside normal operation.

Acceptance criteria: No resource-intensive loop runs while idle. Concurrent jobs stay within the approved limit.

Priority: Essential | Status: Acceptance baseline | Reference: C16, U | Test: AT-OPS-09

Chapter 24

Performance and Quality Targets

These are proposed test-design targets. They are neither current measurements nor contractual guarantees. The proposed baseline is an API with four CPU cores and 8 GB of memory, separate database and vector store, 10 concurrent users, and 1,000 sample documents. Record model type, network, and text volume in the test report.

Target ID	Proposed target	Measurement
PERF-01	API listing and contract p95 at most 2 seconds	Requests after warm-up, excluding model generation time
PERF-02	Waiting feedback within 200 milliseconds of a click	Time from user action to visible UI change
PERF-03	Acceptance and queueing within 3 seconds after transfer and scan	Measure network and scan separately from job registration
PERF-04	First model response content p95 at most 15 seconds	Keep model and network fixed. Count timeouts as failures.
REL-01	Monthly availability target of 99.5%	Agree on maintenance windows and scope of external dependencies.
REC-01	Service recovery within 4 hours and data loss of at most 24 hours	Restore exercise measuring RTO and RPO
QUAL-01	At least 90% of sampled citations actually support the associated claims	Human review of the approved evaluation set
QUAL-02	Zero unauthorized source disclosures in the test set	Negative permission and organization tests. Any incident blocks acceptance.

Note: Report OCR time by file type, page count, scan quality, and provider. A fixed time for all PDFs is unreliable. Token cost shall not be presented as definitive without valid provider pricing. Unknown cost is distinct from zero cost.

Chapter 25

End-to-End Acceptance Scenarios

UC-01 Sign-in and Role Restrictions

Precondition: An active regular account in organization A. Test data must be separate from production data.

Test flow: Sign in, inspect the menu, and try the member URL and API. Record request IDs and observed states.

Expected result: The workspace is usable and administrative operations are denied.

Alternatives: Incorrect password, expired session, deleted account. Related requirements: FR-AUTH and BR.

UC-02 From Text File to Answer

Precondition: An authorized set and a healthy worker. Test data must be separate from production data.

Test flow: Upload a sample TXT, wait for indexing, ask a question with a known answer, and open its source. Record request IDs and observed states.

Expected result: The answer cites a chunk from that document.

Alternatives: Empty file, invalid UTF-8, duplicate file. Related requirements: FR-UP, FR-ING, and FR-RAG.

UC-03 Scanned Document

Precondition: An approved OCR provider and valid quota. Test data must be separate from production data.

Test flow: Upload a scanned PDF, observe the stage, inspect extracted text, and ask a question. Record request IDs and observed states.

Expected result: Text is retrievable and the source belongs to the same document.

Alternatives: OCR off, page limit exceeded, empty output. Related requirements: FR-ING and FR-CHK.

UC-04 File Security Rejection

Precondition: An isolated test environment with an active scanner. Test data must be separate from production data.

Test flow: Submit a controlled malware sample and check the response, queue, and storage. Record request IDs and observed states.

Expected result: Reject before extraction and safely record the reason.

Alternatives: Scanner unavailable or invalid scanner response. Related requirements: FR-UP and SEC.

UC-05 Worker Recovery

Precondition: A long-running job and two test workers. Test data must be separate from production data.

Test flow: Stop the job owner, wait for lease expiry, and resume with the other worker. Record request IDs and observed states.

Expected result: No duplicate effective execution and a consistent final result.

Alternatives: Interruption during upsert and concurrent document deletion. Related requirements: FR-ING and OPS.

UC-06 Query Across All Sets

Precondition: Three sets, with view permission for two. Test data must be separate from production data.

Test flow: Select All, ask a question relevant to all sets, and inspect all sources and context. Record request IDs and observed states.

Expected result: Only the two authorized sets participate.

Alternatives: Permission revoked during conversation. Related requirements: BR and FR-RAG.

UC-07 Cancel an Answer

Precondition: An active request with controlled latency. Test data must be separate from production data.

Test flow: Submit, stop, edit the question, and start a new request. Record request IDs and observed states.

Expected result: The interface becomes usable and the earlier result does not appear under the new request.

Alternatives: Late answer and concurrent network outage. Related requirements: FR-CHAT.

UC-08 Custom Assistant

Precondition: An administrator and authorized sources. Test data must be separate from production data.

Test flow: Set instructions and sources, then start a conversation as an authorized user. Record request IDs and observed states.

Expected result: The answer mode and source scope are honored.

Alternatives: Disabled assistant or unavailable model. Related requirements: FR-AST.

UC-09 Repeated Synchronization

Precondition: A test connector with fixed data. Test data must be separate from production data.

Test flow: Sync, repeat sync, change one source, and sync again. Record request IDs and observed states.

Expected result: Unchanged sources are not duplicated and the change is traceable.

Alternatives: Single-file error, expired lease, invalid credentials. Related requirements: FR-CON.

UC-10 Share and Revoke

Precondition: A conversation with a nonsensitive source. Test data must be separate from production data.

Test flow: Publish to the team and by link, read, add a message, and revoke. Record request IDs and observed states.

Expected result: The snapshot remains fixed, and the revoked link is unusable.

Alternatives: External recipient and tampered token. Related requirements: FR-SHR.

UC-11 Operation Completion Notification

Precondition: A job belonging to a specific user. Test data must be separate from production data.

Test flow: Run successful and failed processing, open the workspace, and read notifications. Record request IDs and observed states.

Expected result: The result reaches the correct user and the read state is stored on the server.

Alternatives: Mark-read error and deleted target. Related requirements: FR-NOT.

UC-12 Persian Report

Precondition: Report data on specified days. Test data must be separate from production data.

Test flow: Choose Persian and a date range, then export Excel and PDF. Record request IDs and observed states.

Expected result: The Solar Hijri calendar and totals match the report.

Alternatives: Midnight boundary, empty range, export failure. Related requirements: FR-AN and DAT.

UC-13 Mobile and Keyboard

Precondition: Real browsers and both languages. Test data must be separate from production data.

Test flow: Test sign-in, drawer, conversation, upload, and modal at reference widths. Record request IDs and observed states.

Expected result: Main operations work without hidden controls or lost focus.

Alternatives: Long filename and mobile keyboard opening. Related requirements: NFR-UI and NFR-A11Y.

UC-14 Disaster Recovery

Precondition: A consistent backup and an independent environment. Test data must be separate from production data.

Test flow: Restore database and files, prepare vectors, and run the document-to-answer flow. Record request IDs and observed states.

Expected result: Ownership, documents, and sources are consistent. Record elapsed time.

Alternatives: Incompatible schema version and incomplete backup file. Related requirements: OPS and DAT.

Chapter 26

Traceability and Delivery Conditions

Each requirement has a corresponding AT identifier in its description. The test owner shall link inputs, environment, result, and evidence. The table shows scenario coverage by topic. Covering a family does not mean that every test in it passed.

Requirement family	Covering scenarios	Primary method
FR-AUTH, FR-USER, and BR	UC-01	Negative API tests and browser tests
FR-UP and FR-KB	UC-02 and UC-04	File integration and access controls
FR-ING and FR-CHK	UC-03 and UC-05	Worker and dependency failures
FR-RAG and FR-CHAT	UC-06 and UC-07	Retrieval, human evaluation, and late requests
FR-AST	UC-08	Assistant contract and source permissions
FR-CON	UC-09	Concurrency and test connection
FR-SHR	UC-10	Snapshot and revocation
FR-NOT	UC-11	Event, notification storage, and SMTP testing
FR-AN and DAT	UC-12 and UC-14	Report reconciliation and data integrity
NFR-UI and NFR-A11Y	UC-13	Real browsers and screen readers
SEC and OPS	UC-04 and UC-14	Security controls, failover, and recovery

Note: Release acceptance requires all mandatory requirements to pass, with explicit approval recorded for exceptions. Cross-organization disclosure, malware acceptance in production, loss of confirmed data, or false processing success blocks acceptance. Prioritize visual defects by their effects on task completion, legibility, and accessibility.

The final test report shall specify code and migration versions, model, OCR provider, browser, and device. Edge, Chrome on Android, and Safari on iPhone shall be recorded as separate environments. Emulating iPhone dimensions in a desktop browser does not replace testing on real Safari.

CI shall run lightweight checks for routine changes. Heavy, load, and OCR tests shall run separately with bounded resources. Concurrency limits and the execution environment shall be configurable. Pipeline results shall show which test groups ran and which did not.

Chapter 27

Open Issues and Decisions

Resolve the following before approving the release baseline. Recording them prevents technical assumptions or conditional capabilities from becoming product commitments.

DEC-01 File Size and Type Limits

Product and development shall align the UI and API allowlists and record boundary tests.

DEC-02 Performance Budget and Capacity

Operations and the product owner shall approve the Chapter 24 targets and reference hardware.

DEC-03 Retention and Deletion

Security and the data owner shall set retention periods for documents, conversations, logs, quarantine, and backups.

DEC-04 Provider Confidentiality

The data owner shall determine which documents may be sent to external model or OCR providers.

DEC-05 Notifications for Regular Users

Approve notification placement and access in the regular user workspace. The current code places the bell in the administrator analytics view.

DEC-06 Confirm Notification Read Status

The current mark-all interface changes local state even after an API error. FR-NOT-03 requires a fix and test.

DEC-07 Central Monitoring and Multi-Instance Alerts

Backend logging and email exist in the code. A shared cooldown and replay behavior must be defined.

DEC-08 Connector Scope

Older architecture documents name web, webhooks, and GitHub. Record connection tests for active SharePoint, Drive, and S3 integrations.

DEC-09 Public Sharing

The product owner and security shall approve unauthenticated links, expiry, and content restrictions.

DEC-10 Legacy Identifier Migration

The test team shall record the mapping from version 2 to version 3 families in the traceability system. This version’s IDs form the new detailed baseline.

Chapter 28

References and Glossary

28.1 References

R01 - ISO/IEC/IEEE 29148:2018, requirements engineering. Reference for requirements structure and information. No formal conformity assessment is claimed.

R02 - W3C Web Content Accessibility Guidelines 2.2, the reference for interface accessibility targets. https://www.w3.org/TR/WCAG22/

Code references:

• C01 - backend/app/models, backend/app/api/routes, and documents/architecture/ARCHITECTURE.md

• C02 - backend/app/api/routes/auth.py and backend/app/core/config.py

• C03 - backend/app/api/routes/users.py and frontend/app/src/pages/UsersPage.tsx

• C04 - backend/app/api/routes/document_sets.py and frontend/app/src/pages/UploadFilesPage.tsx

• C05 - backend/app/services/documents/upload_security.py and backend/app/core/config.py

• C06 - backend/app/services/documents/document_jobs.py, backend/app/services/documents/cloud_ocr.py, and backend/app/services/operations/worker_heartbeat.py

• C07 - backend/app/services/documents/text_chunker.py and backend/app/services/documents/semantic_chunker.py

• C08 - backend/app/services/retrieval/retrieval.py, backend/app/api/routes/rag.py, backend/app/api/routes/research.py, and backend/app/api/routes/search.py

• C09 - backend/app/api/routes/conversations.py, frontend/app/src/features/chat/components/ChatInput.tsx, and frontend/app/src/features/chat/components/ChatWindow.tsx

• C10 - backend/app/api/routes/assistants.py and backend/app/schemas/assistant.py

• C11 - backend/app/services/connectors/connector_scheduler.py, backend/app/services/connectors/connector_sync.py, and backend/app/services/connectors/connector_lock.py

• C12 - backend/app/api/routes/chat_shares.py and backend/app/api/routes/feedback.py

• C13 - backend/app/api/routes/analytics.py and frontend/app/src/pages/AnalyticsPage.tsx

• C14 - backend/app/api/routes/notifications.py, backend/app/services/operations/operational_alerts.py, and frontend/app/src/features/notifications/components/NotificationCenter.tsx

• C15 - documents/development/DESIGN.md and interface language and theme settings

• C16 - backend/app/main.py, backend/app/services/operations/operational_metrics.py, and backend/app/services/operations/operational_alerts.py

28.2 Glossary

Term	Meaning
RAG	Generating an answer using context retrieved from sources
OCR	Extracting text from an image or scanned document
Chunk	A text fragment used for retrieval and citation
Embedding	A numerical representation of text for comparison and retrieval
Grounded	Supported by verifiable evidence in sources
Tenant	An independent organization that defines a data and authorization boundary
Lease	Temporary ownership of a job with a defined expiration
Heartbeat	A periodic signal that a worker or job owner is alive
Snapshot	A fixed version of content at a specific time
Idempotency	Repeating a request without unintended duplicate effects
RTO	Target maximum time to restore service
RPO	Maximum acceptable time window of data loss
95th percentile	A time at or below which 95% of samples fall